

RAOP: Runtime-Adaptive Operator Selection for Routability-Driven Placement

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Abstract

Routability-driven placement often applies one response rule despite design-dependent congestion morphology. RAOP instead uses the first global-routing state to select among four response policies with different physical actions and admissibility rules; unresolved states receive one rollback-capable scalar probe. Five heuristic thresholds are fixed before downstream evaluation. On 20 compatibility-processed ISPD 2015 instances, RAOP reduces detailed-routing DRC by 27.49% relative to fixed legacy inflation and by 73.82% relative to no inflation on a paired geometric-mean basis. Fixed-policy controls, a static selector, and branch knockouts characterize the contribution of the selected paths. We also evaluate the frozen selector on eight compatibility-processed ICCAD 2017 instances, where only scalar and directional XY are activated. In an equal-route-opportunity control, both RAOP and fixed legacy receive 15 route-optimization rounds and 16 global-routing calls per design. RAOP gives a 4.20% lower DRC aggregate, with a 29.24% higher detailed-routing time aggregate. These results show that response-policy selection can improve routability, but its benefit and cost remain design dependent.

Keywords: VLSI placement, routability optimization, global routing, detailed routing, congestion mitigation, cell inflation, runtime operator selection

1. Introduction

Global placement constrains the solution space available to legalization and routing. Wirelength and density are necessary placement objectives, but they do not directly encode the design-rule violations (DRCs) that emerge after detailed routing. Routability-driven placers therefore incorporate congestion estimation, routing feedback, whitespace redistribution, or rule-aware refinement before committing the downstream flow [1–5].

An additional control question considered here is not only how strongly to inflate cells, but which physical response operator is appropriate for the observed routing state. We hypothesize that congestion states with different concentration, directional imbalance, and short-term response may call for different admissible physical responses. A single fixed operator can conflate these mechanisms and may be effective for one congestion pattern but unsuitable for another.

Existing routability-driven methods express state-dependent responses through whitespace allocation, cell or net inflation, congested-object movement, cluster spreading, congestion-aware forces, learned maps, continuous inflation/shrinking ratios, or dynamic objective weights [6–17]. RAOP asks a different control question: whether first-route morphology and a short conditional response can select the physical action itself. We study this architecture with fixed-policy controls and branch ablations on compatibility-processed ISPD 2015, then evaluate the frozen selector and a matched-opportunity control on compatibility-processed ICCAD 2017.

The primary contribution is the RAOP selector architecture. Let a first-route observation be s_1 and let the available operator family be

$$\mathcal{O} = \{o_{\text{legacy}}, o_{\text{xy}}, o_{\text{nd}}, o_{\text{scalar}}\}. \quad (1)$$

RAOP uses a hierarchical rule: it selects an operator directly when a morphology gate fires, and otherwise requests one response observation before the final operator selection. Here an operator denotes a complete response policy: cell-level actuation together with its admissible area updates, probe-retention behavior where applicable, termination rule, and candidate-selection key. The four operators therefore differ in update direction, geometry, admissible area changes, and control semantics.

The contributions are as follows.

1. We define a runtime selector architecture that maps first-global-route morphology, and when needed one rollback-capable response observation, to one of four heterogeneous physical/admissibility response policies.
2. We evaluate the complete system, fixed responses, branch knockouts, and local decision stability on 20 compatibility-processed ISPD 2015 instances.
3. We report frozen ICCAD 2017 behavior and an equal-route-opportunity control that accounts for the routing calls used to acquire selector evidence.

The broader XPlace and route-aware studies provide context for the underlying flow [18, 19].

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2. Related Work

2.1. Fixed Response Mechanisms

Congestion-driven placement has long created whitespace around routing pressure through cell inflation [6]. Later methods integrated routing-demand estimation, analytical placement, spreading, or explicit whitespace allocation [20–22]. These approaches establish the main physical responses used by routability-driven placement: change cell area, redistribute whitespace, and alter the local density field around congested regions.

Methods that couple routing feedback to a prescribed refinement schedule include SimPLR, which couples probabilistic routing with placement refinement [2]; Ripple and Ripple 2.0, which combine router feedback with movement and staged spreading [7, 23, 24]; and rule-aware placers that incorporate technology, hierarchy, region, and directional constraints [4, 25]. RePlAce supplies an electrostatic global-placement formulation with routability validation [5], while XPlace supplies the extensible GPU engine used here [18]. Net-distribution methods perform directional net inflation and replacement [8]; global/local methods combine net penalties and cluster inflation [9]; PPOM reallocates whitespace after global placement [26]; and later approaches add routing-path forces, differentiable movement, and pin-access mitigation [10, 19]. Across the methods reviewed here, adaptation is typically expressed through a response coefficient, magnitude, objective weight, prediction target, or refinement schedule. RAOP focuses instead on selecting the admissible physical action at runtime.

2.2. Adaptive Magnitude and Parameter Control

PUFFER provides a strategy-oriented precedent: its SMBO/TPE procedure explores continuous parameters and optional padding strategies during method development, and the reported configuration is then applied to other designs [27]. RUPlace alternates placement and routing while computing continuous inflation or shrinking ratios [16]; C3PO dynamically balances timing, routability, and wirelength objective weights during placement [17]. Prediction- and learning-guided approaches include RouteNet, PROS, reinforcement-learning placement-parameter search, differentiable congestion objectives, and LACO [11–14, 28]; explainable multi-task guidance combines congestion and wirelength impact inside one inflation formula [15].

These studies establish adaptation in routability-driven placement, primarily through parameters, objective weights, or response magnitudes within a prescribed mechanism. RAOP instead selects among policies with different actuation and admissibility. This distinction is related to per-instance algorithm selection, which maps descriptive features to a member of an algorithm portfolio [29]. To the best of our knowledge, prior routability-driven placers do not combine design-level first-route morphology, a conditional response observation, and selection among these heterogeneous response policies.

2.3. Behavioral Probing and Comparison Budgets

Short solver trajectories can serve as behavioral features for per-instance selection [30]. RAOP uses its tentative scalar response in this role. Because obtaining that response requires another global-routing call, its acquisition cost belongs in the comparison budget; feature-budget analysis in algorithm selection motivates the same accounting principle [31]. We therefore report route-optimization rounds and global-routing calls separately from placement compute, detailed-routing time, work per call, and trajectory state. The equal-route-opportunity control matches the first two quantities.

3. Method

3.1. Problem Formulation and Intervention Point

Let x^0 be the placement at the first routing trigger and let $\mathcal{R}(x)$ return global-routing congestion fields and metrics. RAOP observes

$$s_1 = (\mu_C, \rho_{1\%}, \rho_{\text{conn}}, \eta_{HV}) \quad (2)$$

from $\mathcal{R}(x^0)$. The selector applies two ordered morphology gates to determine whether direct operator selection is possible; only unresolved states receive a one-step probe that evaluates a response statistic g_{probe} after a tentative scalar update. Innovus detailed routing is downstream evaluation only and is absent from both s_1 and g_{probe} .

After selection, each operator supplies cell-level actuation through its update direction, geometry rule, candidate-selection key, and applicable termination rule. The thresholds, descriptor definitions, and operator constants are the flow-specific settings evaluated in this study.

Figure 1 shows the control flow used in this study. The strong-hotspot gate is evaluated first and maps concentrated high-mass congestion to legacy inflation; selection of the directional XY operator is considered only when that condition is not satisfied. Response information is requested only for unresolved states.

3.2. First-Route State Descriptor

Let D_{ij} and K_{ij} be the aggregate routing demand and capacity in global-routing bin (i, j) after summing the routing layers used by the placement estimator. Define the valid scalar-bin set $\mathcal{B} = \{(i, j) : K_{ij} > 0\}$ and $N_b = |\mathcal{B}|$. Zero-capacity bins are excluded from the scalar descriptor sums and order statistics below. The retained aggregate capacities satisfy $K_{ij} > 0$ in every bin of the evaluated routing maps, so $\mathcal{B}$ is the complete two-dimensional routing grid in the reported experiments. The scalar congestion field is the nonnegative overflow ratio

$$C_{ij} = \max\left(\frac{D_{ij}}{K_{ij}} - 1, 0\right), \quad (3)$$

for $(i, j) \in \mathcal{B}$. Thus C_{ij} is an overflow field rather than raw demand/capacity utilization. Every unqualified bin sum in this subsection is over $\mathcal{B}$. Mean congestion is

$$\mu_C = \frac{1}{N_b} \sum_{i,j} C_{ij}. \quad (4)$$

Table 1: Structural comparison of nearby adaptive methods by adaptation unit, response space, routing feedback, and budget treatment.

Method	Adaptation unit	Response/action space	Routing feedback, probing, and budget treatment
PUFFER [27]	Method-development search	Continuous parameters and optional padding strategies	SMBO/TPE search during method development; the reported configuration is subsequently applied to other designs
RUPlace [16]	Alternating placement–routing iterations	Continuous cell inflation or shrinking ratio	Online routing feedback adjusts response magnitude within one operator family
C3PO [17]	Placement iterations	Dynamic timing/routability/wirelength objective weights	Online state changes adjust the objective balance
Xing et al. [15]	Learned multi-task guidance	Congestion and wirelength influence combined in one explainable inflation formula	Learned guidance acts within a prescribed inflation formulation
RAOP	Design-level first route, plus a conditional second route	Selection among monotone isotropic, directional, no-deflation, and bidirectional scalar policies	A short response is acquired only for unresolved states; its route opportunity is counted as part of the comparison budget

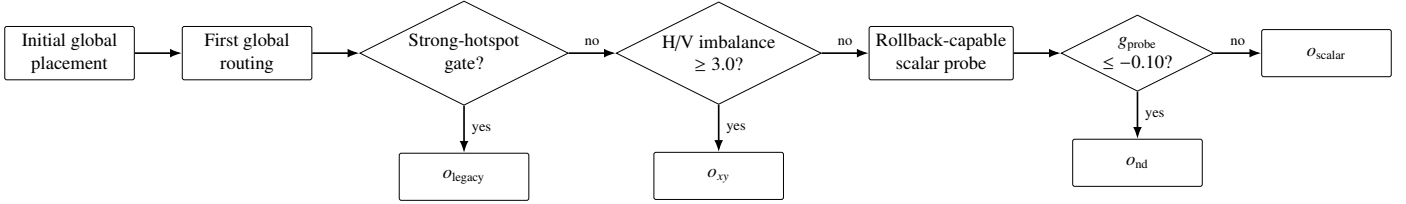


Figure 1: Runtime response-operator selection from first-route morphology. The probe is executed only for unresolved first-route states. Detailed routing is not a selector input.

With $k = \max(1, \lceil 0.01N_b \rceil)$ and $\mathcal{T}_k$ the k bins with largest C_{ij} , top-bin concentration is

$$\rho_{1\%} = \frac{\sum_{(i,j) \in \mathcal{T}_k} C_{ij}}{\sum_{i,j} C_{ij}}. \quad (5)$$

Let $M_C = \sum_{i,j} C_{ij}$. When $M_C = 0$, the implementation defines both $\rho_{1\%} = 0$ and $\rho_{\text{conn}} = 0$; the positive-bin set and hot-component set are empty. If both directional field means are also zero, $\eta_{HV} = 1$. The global router also provides an estimated-short cost $S(R) = C_{\text{wire}}(R) + C_{\text{via}}(R)$ from its capacity, wire-demand, and via-demand maps. The two terms are the layer-scaled positive wire-overflow area and the via demand at routing locations where wire demand exceeds capacity, respectively. A clean first route has zero overflow nets and $S(R_1) \leq 10^{-6}$. In that case the implementation disables cell inflation and returns the first-route candidate before any cell-geometry update or follow-up global-routing call. A zero-mass field that does not satisfy this short-cost criterion remains an unresolved state and follows the ordinary probe path.

A hot mask uses threshold

$$\tau_{\text{hot}} = \begin{cases} \max(0.05, Q_{0.90}(\{C_{ij} : C_{ij} > 0\})), & M_C > 0, \\ 0.05, & M_C = 0. \end{cases} \quad (6)$$

where $Q_{0.90}$ denotes the 90th percentile of the nonempty positive-bin set. Four-neighbor connected components are formed on positive bins satisfying $C_{ij} \geq \tau_{\text{hot}}$. If $\mathcal{K}_{\text{max}}$ is the component with largest congestion mass, then

$$\rho_{\text{conn}} = \frac{\sum_{(i,j) \in \mathcal{K}_{\text{max}}} C_{ij}}{\sum_{i,j} C_{ij}}. \quad (7)$$

The conjunction of μ_C , $\rho_{1\%}$, and ρ_{conn} prevents an isolated peak or a diffuse weak field from triggering the strong-hotspot gate.

The directional fields are formed analogously on $\mathcal{B}_d = \{(i, j) : K_{ij}^d > 0\}$ as $C_{ij}^H = \max(D_{ij}^H/K_{ij}^H - 1, 0)$ and $C_{ij}^V = \max(D_{ij}^V/K_{ij}^V - 1, 0)$, where demand and capacity are separately aggregated over horizontal- and vertical-preferred routing layers. Zero directional-capacity bins are excluded from the corresponding field mean. For field means $\overline{C}^H$ and $\overline{C}^V$,

$$\eta_{HV} = \frac{\max(\overline{C}^H, \overline{C}^V)}{\min(\overline{C}^H, \overline{C}^V)}. \quad (8)$$

If the denominator is zero, the implementation returns infinity when the numerator is positive and one when both are zero.

For an unresolved state, the algorithm snapshots positions, sizes, pin offsets, density state, and scheduler state; executes one scalar momentum update; and reruns global routing. If O_1 and O_2 are the numbers of overflow nets before and after this probe,

$$g_{\text{probe}} = \frac{O_1 - O_2}{\max(O_1, 1)}. \quad (9)$$

A negative value denotes deterioration. If no-deflation is selected, the complete snapshot is restored and that operator restarts from the pre-probe state. If scalar is selected, the probe is retained as its first update.

3.3. Threshold Selection and Hierarchical Selector

The five selector thresholds are $\theta = (0.010, 0.20, 0.10, 3.0, -0.10)$. They are flow-specific heuristic

settings, manually chosen as rounded values from the observed first-global-route descriptor and probe-response scales and fixed before downstream evaluation; final DRC is not used to fit them. The additional 0.05 floor and 90th-percentile operation in τ_{hot} are descriptor-construction constants, not additional selector branches.

The selector selects o_{legacy} when

$$\mu_C \geq 0.010 \wedge \rho_{1\%} \geq 0.20 \wedge \rho_{\text{conn}} \geq 0.10. \quad (10)$$

If that conjunction fails, $\eta_{HV} \geq 3.0$ selects o_{xy} . The remaining unresolved instances are probed; $g_{\text{probe}} \leq -0.10$ selects o_{nd} , otherwise o_{scalar} . Figure 2 shows the observed selector quantities relative to their boundaries, and Figure 3 gives executable-level pseudocode.

3.4. Response Operators and Shared Safeguards

The legacy operator is the original monotone congestion-driven inflation path. At counted routing round t , it forms an isotropic inflation field from $(C^t + 1)^2$, clamps the field to $[1, 2]$, projects it onto cells by overlap, and applies no deflation.

The other three operators share a momentum controller. Let $A_{k,ij}^t$ be the overlap area between cell k 's current analytical footprint and routing bin (i, j) , and let A_k^t be that footprint's area. The implementation projects any bin field F to the cell by

$$\Pi_k^t(F) = \frac{1}{A_k^t} \sum_{i,j} A_{k,ij}^t F_{ij}. \quad (11)$$

At the start of counted momentum update t , C^t denotes the current routing field from the latest global route; the first-route descriptor uses $C = C^1$. The update computes its stimulus from C^t , changes the temporary analytical geometry, and is followed by placement refinement and global routing that produce C^{t+1} . The aggregate capacity grid and hence $\mathcal{B}$ are fixed across these updates, and C^t is extended by zero outside $\mathcal{B}$ only for cell projection. Define $c_k^t = \Pi_k^t(C^t)$, and let μ_C^t and σ_C^t be the mean and population standard deviation of that same current field over $\mathcal{B}$. The denominator guard ϵ is the machine epsilon of the local-congestion tensor's floating-point data type. The stimulus and momentum update are

$$u_k^t = \text{clip}\left(\frac{c_k^t - \mu_C^t}{\max(\sigma_C^t, \epsilon)}, -1, 1\right), \quad (12)$$

$$\Delta r_k^t = 0.4 \Delta r_k^{t-1} + 0.6 u_k^t, \quad (13)$$

$$r_k^t = \text{clip}(r_k^{t-1} + \Delta r_k^t, r_{\min}, 2.0). \quad (14)$$

The controller starts from the original geometry with $r_k^0 = 1$ and $\Delta r_k^0 = 0$. For o_{scalar} and o_{xy} , $r_{\min} = 0.9$ and negative updates are legal. For o_{nd} , $r_{\min} = 1.0$ and Δr_k^t is projected onto the non-negative half-line. Positive area change is limited by available whitespace; negative change is applied before the remaining positive budget is computed. Filler area preserves target-area accounting, and pin offsets follow cell geometry.

Let $s_{x,k}^0$ and $s_{y,k}^0$ be cell k 's original analytical width and height. Before application of the density-proxy clamp, the scalar and no-deflation operators set

$$s_{x,k}^t = s_{x,k}^0 \sqrt{r_k^t}, \quad s_{y,k}^t = s_{y,k}^0 \sqrt{r_k^t}. \quad (15)$$

Thus r_k^t is an area ratio relative to the original geometry, rather than a multiplier applied repeatedly to the current dimensions. The directional operator preserves the requested area-scaling product but divides each step between width and height. For $d \in \{H, V\}$, define $\mathcal{B}_d^t = \{(i, j) : K_{ij}^{d,t} > 0\}$ and $U_{ij}^{d,t} = D_{ij}^{d,t} / K_{ij}^{d,t}$ on $\mathcal{B}_d^t$. Both $\mathcal{B}_H^t$ and $\mathcal{B}_V^t$ are nonempty in every evaluated routing map. For projection, $U^{d,t}$ is extended by zero outside $\mathcal{B}_d^t$, giving $\ell_{k,d}^t = \Pi_k^t(U^{d,t})$; zero-capacity bins are excluded from directional field means and order statistics. Let $H_{k,d}^{t-1}$ be its directional history. The operator $N_{0.95}$ divides positive active entries by their 95th percentile and clips them at one; if no positive active entry exists, it returns the all-zero field. Otherwise zero entries remain zero and the percentile denominator has an ϵ floor. Define

$$p_{k,d}^t = N_{0.95}([\ell_{k,d}^t - 1]_+), \quad (16)$$

$$n_{k,d}^t = N_{0.95}([H_{k,d}^{t-1} - \ell_{k,d}^t - 0.10]_+), \quad (17)$$

where $n_{k,d}^t$ is active only when $H_{k,d}^{t-1} \geq 1.05$, $\ell_{k,d}^t \leq 0.90$, the utilization drop is at least 0.10, and the scalar update requests deflation. The global bias uses a routing-congestion summary distinct from the directional field means in η_{HV} . Let $N_b^{d,t} = |\mathcal{B}_d^t| \geq 1$ and sort $Z_{ij}^{d,t} = \max(U_{ij}^{d,t}, 1)$ over $\mathcal{B}_d^t$ as $Z_{(1)}^{d,t} \geq \dots \geq Z_{(N_b^{d,t})}^{d,t}$. For $\mathcal{A} = \{0.005, 0.01, 0.02, 0.05, 0.10, 0.20, 0.50\}$ and $m_a^{d,t} = \lfloor a N_b^{d,t} \rfloor + 1$, define

$$R_d^t(a) = \frac{1}{m_a^{d,t}} \sum_{q=1}^{m_a^{d,t}} Z_{(q)}^{d,t}, \quad \bar{R}_d^t = \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} R_d^t(a). \quad (18)$$

With $e_d^t = [\bar{R}_d^t - 1]_+$ and $g^t = (e_H^t - e_V^t) / (e_H^t + e_V^t)$ (set to zero when $e_H^t + e_V^t = 0$), the inflation and deflation biases are

$$b_k^{+,t} = 0.75(p_{k,H}^t - p_{k,V}^t) + 0.25 g^t \max_d p_{k,d}^t, \quad (19)$$

$$b_k^{-,t} = n_{k,H}^t - n_{k,V}^t. \quad (20)$$

The controller uses $b_k^t = \text{clip}(b_k^{-,t}, -1, 1)$ on a deflation step and $b_k^t = \text{clip}(b_k^{+,t}, -1, 1)$ otherwise, then sets

$$w_{x,k}^t = \text{clip}(0.5 + 0.20 b_k^t, 0.35, 0.65), \quad w_{y,k}^t = 1 - w_{x,k}^t. \quad (21)$$

Let $\gamma_{q,k}^t$ be the controller's cumulative analytical size factor in direction $q \in \{x, y\}$, so $s_{q,k}^t = s_{q,k}^0 \gamma_{q,k}^t$ before application of the density-proxy clamp and $\gamma_{q,k}^0 = 1$. Its provisional values and bounded reconstruction are

$$\tilde{\gamma}_{x,k}^t = \gamma_{x,k}^{t-1} \left(\frac{r_k^t}{r_k^{t-1}} \right)^{w_{x,k}^t},$$

$$\tilde{\gamma}_{y,k}^t = \gamma_{y,k}^{t-1} \left(\frac{r_k^t}{r_k^{t-1}} \right)^{w_{y,k}^t},$$

$$a_k^t = \text{clip}(\tilde{\gamma}_{x,k}^t / \tilde{\gamma}_{y,k}^t, 0.8, 1.25), \quad (\gamma_{x,k}^t, \gamma_{y,k}^t) = (\sqrt{r_k^t a_k^t}, \sqrt{r_k^t / a_k^t}). \quad (22)$$

Hence $\gamma_{x,k}^t \gamma_{y,k}^t = r_k^t$; the bound is on scale factors, not physical cell aspect. History starts at $H_{k,d}^0 = 0$ and updates as $H_{k,d}^t = \max(0.95 H_{k,d}^{t-1}, \ell_{k,d}^t)$.

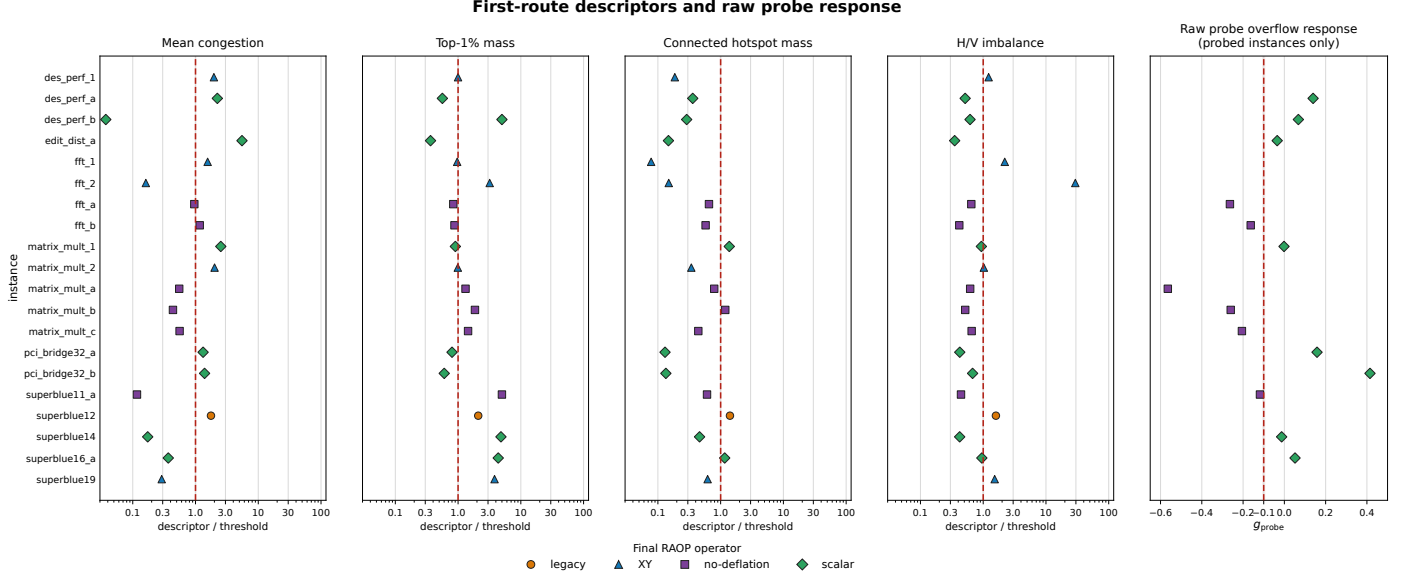


Figure 2: First-route selector descriptors and raw probe response. The first four panels show the first-global-route descriptors for all 20 benchmark instances, normalized by their corresponding selector thresholds; the dashed line at 1 marks each boundary, and the three hotspot descriptors are interpreted jointly for the strong-hotspot gate. The fifth panel shows the raw probe response g_{probe} only for instances reaching the probe stage, with the dashed line at -0.10 ; it is not normalized by a threshold. Colors and marker shapes indicate the final RAOP operator selection rather than the outcome of the individual descriptor shown in each panel. A single descriptor crossing does not by itself determine the final selection because the gates are ordered and the hotspot descriptors are conjunctive.

All reported configurations enable the same density-proxy clamp for every inflation path. Let δ_q be the placement-density-bin side length in direction $q \in \{x, y\}$ and define the common absolute density-proxy lower bound $m_q = \sqrt{2}\delta_q$. For scalar and no-deflation, $s_{q,k}^t$ is the analytical dimension defined by Eq. (15). For legacy, it is the pre-clamp analytical dimension produced by the projected legacy inflation update. In these three paths, the density solver receives $\bar{s}_{q,k}^t = \max(s_{q,k}^t, m_q)$. For XY, the same bound is enforced after directional shaping through

$$\begin{aligned} \bar{s}_{q,k}^t &= \max(s_{q,k}^0 \sqrt{r_k^t}, m_q), \\ z_{q,k}^t &= \bar{s}_{q,k}^t \frac{\gamma_{q,k}^t}{\sqrt{r_k^t}}, \quad \chi_k^t = \max\left(1, \frac{m_x}{z_{x,k}^t}, \frac{m_y}{z_{y,k}^t}\right), \\ \bar{s}_{q,k}^t &= \chi_k^t z_{q,k}^t. \end{aligned} \quad (23)$$

The density expansion weight is $\alpha_k^t = s_{x,k}^t s_{y,k}^t / (\bar{s}_{x,k}^t \bar{s}_{y,k}^t)$, so the enlarged proxy support retains the temporary analytical area. The relations above for r_k^t and $\gamma_{q,k}^t$ therefore describe pre-clamp geometry; the clamp is density-bin-derived rather than relative to original cell size.

The resized widths/heights, filler sizes, and rescaled pin offsets are temporary analytical variables used by global-placement density and routing-feedback calculations; they do not modify the LEF masters or the original cell-size array stored by the placement database. At the end of route-driven global placement, the implementation restores the optimizer-facing movable-size tensor and original pin offsets, total-movable-area scalar, and target/init-density state, then selects a recorded candidate by center coordinates. The default legalization and detailed-placement stages receive those coordinates together

with the original raw/global placement databases. When a DEF is written, cell lower-left locations are recomputed from the selected centers using the original database cell sizes, the locations are committed to the global-placement database, and that database writes the placement. Consequently, every DEF routed by Innovus contains original legal LEF/master geometry; temporary analytical resizing and pin-offset changes are not serialized.

All operators share finite area and iteration safeguards. For candidate j and component $x \in \{\text{wire}, \text{via}\}$, the legacy and no-deflation selectors use $\bar{C}_{x,j} = C_{x,j} / \max(C_{x,0}, 10^{-12})$, where $j = 0$ is the first recorded candidate. They try the wire-weight sequence $(0.93, 0.75, 0.55, 0.30, 0.18, 0)$ and, for each weight λ in order, lexicographically minimize

$$(\lambda \bar{C}_{\text{wire},j} + (1 - \lambda) \bar{C}_{\text{via},j}, N_{\text{ovfl},j}, (\bar{R}_{H,j} + \bar{R}_{V,j})/2). \quad (24)$$

Here $N_{\text{ovfl},j}$ is candidate j 's overflow-net count, and $\bar{R}_{d,j}$ is the directional routing-congestion summary defined above for that candidate. The first minimizer satisfying $C_{\text{via},j} \leq 1.01 \min_l C_{\text{via},l}$ is returned. Their numerical budgets, geometry bound, convergence tolerances, probe accounting, and the other candidate-selection keys are listed in Supplementary Table S1; these settings are fixed across every instance and experiment.

4. Experimental Methodology

4.1. Benchmark Provenance and Compatibility Processing

The original design family is the ISPD 2015 blockage-aware detailed-routing-driven placement benchmark, whose fence regions, placement/routing blockages, density constraints, and revised floorplans were introduced for detailed-routing-oriented

Require: First routing-trigger placement x^0 ; global router $\mathcal{R}$; selector thresholds θ ; maximum counted operator-update rounds $T = 15$

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1:  $R_1 \leftarrow \mathcal{R}(x^0)$ ; compute  $s_1 = (\mu_C, \rho_{1\%}, \rho_{\text{conn}}, \eta_{HV})$ , using  $\rho_{1\%} = \rho_{\text{conn}} = 0$  when  $\sum C = 0$ 
2: Initialize the candidate pool with  $x^0$  and  $R_1$ ;  $n \leftarrow 0$ 
3: if  $O_1 = 0$  and  $S(R_1) \leq 10^{-6}$  then
4:   Disable cell inflation and return  $x^0$ ; do not apply an operator update or rerun global routing
5: end if
6: if  $\mu_C \geq 0.010$  and  $\rho_{1\%} \geq 0.20$  and  $\rho_{\text{conn}} \geq 0.10$  then
7:    $o \leftarrow o_{\text{legacy}}$ 
8: else if  $\eta_{HV} \geq 3.0$  then
9:    $o \leftarrow o_{xy}$ 
10: else
11:   Snapshot placement, geometry, pins, density, and scheduler state as  $q^0$ 
12:   Apply one scalar momentum update; obtain  $R_2 \leftarrow \mathcal{R}(x^1)$ 
13:    $g_{\text{probe}} \leftarrow (O_1 - O_2) / \max(O_1, 1)$ 
14:   if  $g_{\text{probe}} \leq -0.10$  then
15:     Restore  $q^0$ ;  $o \leftarrow o_{\text{nd}}$ ;  $n \leftarrow 0$ 
16:   else
17:     Retain the probe state;  $o \leftarrow o_{\text{scalar}}$ ;  $n \leftarrow 1$ 
18:     Append retained  $x^1$  and  $R_2$  to the candidate pool
19:   end if
20: end if
21: while  $n < T$  and the termination rule of  $o$  is not reached do
22:   Apply the update admissible under  $o$  using the current routing field; enforce area and geometry safeguards
23:   Run global routing to form the next routing state; record the candidate placement and operator-specific candidate-selection key
24:    $n \leftarrow n + 1$ 
25: end while

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Ensure: Best recorded placement according to the selected operator's candidate-selection key

Figure 3: Pseudocode of the complete RAOP control path. The clean-route check terminates before physical-response actuation. The retained scalar probe counts as the first of T counted operator-update rounds; a probe rolled back before no-deflation does not count toward the restarted no-deflation budget. A branch knockout replaces only the selected branch by o_{scalar} .

evaluation [32, 33]. The evaluated suite contains compatibility-processed versions of all 20 benchmark instances. First-route descriptor and probe-response scales from this processed family informed the fixed heuristic thresholds, so we use the suite for system development and within-flow comparison. All four policies are exercised, enabling complete-system and branch-ablation analysis.

The compatibility procedure applies representation normalization, routing-constraint compatibility transformation, and manufacturing-grid corrections to the ISPD 2015 inputs. Where the placement engine cannot represent a fence-region construct, the corresponding unsupported constraint is removed during this preparation; this occurred for 9 of 20 instances, marked by a dagger in Table 2. Every matched variant receives the same prepared instance. Detailed transformation records are provided in Supplementary Tables S2 and S3. Because the inputs are processed, the resulting values are intended for matched comparisons within this suite, not as official ISPD 2015 contest scores or direct comparisons with results from unmodified contest inputs.

The external evaluation uses eight instances from the IC-CAD 2017 multi-deck standard-cell legalization benchmark [34]. Its compatibility preparation merges technology and

cell LEFs, removes REGIONS/GROUPS fence constructs that the placement engine cannot represent, and aligns FIXED/COVER coordinates to the manufacturing grid. Every matched variant receives the same prepared input. Because preprocessing removes legalization constraints, the results are comparisons within this placement-to-routing substrate rather than official contest scores.

4.2. Implementation and Routing Environment

Detailed-placement DEFs are evaluated with Cadence Innovus release v22.30-s003_1 and NanoRoute on CentOS Linux 7.9 using an Intel Xeon Gold 6430 host with 32 CPU cores and 178 GB memory. The Innovus detailed-routing flow uses Metal1–Metal9 and 8 CPUs, and enables congestion analysis, post-route via swapping, wire spreading, and via optimization. The primary endpoint is final detailed-routing DRC count; detailed-routing wirelength (DRWL), via count, placement time (PT), and detailed-routing time (RT) are secondary descriptive outcomes. Each reported DEF was routed independently three times. DRWL, via count, and DRC were identical across the three runs; RT is reported as the arithmetic mean over those runs. PT is taken from the corresponding placement run. PT and RT are used only as descriptive runtime measures.

The equal-route-opportunity control uses the same Innovus environment and routing configuration as the ISPD 2015 flow.

4.3. Matched Experimental Variants

Eight variants share the same 20 benchmark instances and downstream flow. E0 is no inflation. E1 preserves the original XPlace-Route legacy inflation mechanism without runtime operator selection. E2 is the fixed-scalar control. E3 is a static morphology selector that applies the legacy and XY gates and sends all unresolved instances directly to scalar without probing. E4 is RAOP, the complete runtime-adaptive operator selector. E5a, E5b, and E5c remove the legacy, XY, and no-deflation branches, respectively, and redirect affected instances to scalar. In the following, legacy, XY, no-deflation, and scalar denote the four response operators, whereas RAOP denotes the selector that chooses among them. All operator constants are fixed across instances and experiments; none is tuned per instance.

The matched contrasts are E4–E0 for the complete system relative to no inflation; E4–E1 and E4–E2 for the original fixed legacy and fixed scalar systems; E4–E3 for the added probe-conditioned path after the static gates; and E4–E5a/b/c for replacing each selected non-scalar branch by scalar. Because the response paths differ in probe retention, legal deflation, stopping rules, and candidate selection, these are complete-system comparisons. The external matrix retains E0, E1, and E4.

4.4. Frozen External Evaluation Protocol

The selector, five thresholds, response operators, and baseline definitions were frozen before the ICCAD evaluation; preparation was limited to compatibility processing and dataset integration. Both ICCAD campaigns used seed 0 and deterministic DEF generation. Final Innovus DRC is the downstream endpoint and never a selector input. The ICCAD evaluation therefore assesses the frozen selector on a second design family.

4.5. Equal-Route-Opportunity Control

The control compares fixed legacy E1-EQ with frozen RAOP E4-EQ on all eight compatibility-processed ICCAD 2017 instances. A counted opportunity round consists of one placement-refinement stage followed by one global-routing evaluation; Route Iter denotes the corresponding logged route-optimization event. The initial trigger route precedes these rounds and is not counted, but it is retained as an evaluated global-routing candidate. Both variants are assigned a common horizon of $T = 15$ counted opportunity rounds. If an operator’s ordinary termination condition is reached before round T , further cell-size actuation is disabled and its current analytical size state is frozen; placement refinement and the associated global-routing evaluation continue for the remaining opportunity rounds. The global-routing evaluation associated with round 15 serves as the final candidate-selection evaluation. Together with the initial trigger route, this yields 15 counted Route Iter events and 16 global-routing evaluations and candidates for every variant–design pair. E4-EQ selects scalar

on six instances and directional XY on two; legacy and no-deflation are not exercised. Selector decisions, pre-freeze actuation, ordinary termination conditions, and candidate-selection keys remain unchanged; only continuation after early termination differs from the ordinary campaign. The control therefore evaluates scalar and XY under matched route opportunity. Detailed experiment provenance is provided in Supplementary Table S4.

The within-campaign estimand is the paired E4-EQ/E1-EQ outcome under matched opportunity counts. Operator semantics can still change placement work, state trajectories, routed difficulty, and work per routing call.

4.6. Aggregation and Paired Interpretation

For experiment G , reference E4, design set $\mathcal{D}$, and lower-is-better metric m , the ratio tables report

$$R_G(m) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \frac{m_G(d)}{m_{E4}(d)}. \quad (25)$$

Here $|\mathcal{D}| = 20$ for Tables 2 and 3, and $|\mathcal{D}| = 8$ for Table 5. Thus E4 is normalized to 1.00 and a value above one indicates a larger metric than E4. This is the arithmetic mean of matched per-instance ratios, not a ratio of raw means.

All DRC counts used in the paired comparisons are positive, so the paired geometric ratio of E4 to reference B is

$$G_{E4/B} = \exp\left(\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \ln \frac{\text{DRC}_{E4}(d)}{\text{DRC}_B(d)}\right), \quad (26)$$

and the reported reduction is $100(1 - G_{E4/B})\%$. W/T/L denotes the number of instances on which E4 has lower/equal/higher DRC than the named reference. Affected-only branch statistics use the same definitions on the instances whose allocation changes. No population p -values are reported.

For the equal-route-opportunity control, the same formula is applied to E4-EQ/E1-EQ for DRC, DRWL, vias, PT, and the per-DEF arithmetic-mean detailed-routing time. The median of the eight percentage changes and W/T/L are reported beside each aggregate. Leave-one-design-out geometric ratios measure influence on the frozen DRC aggregate; they are not alternate primary endpoints.

4.7. Threshold-Stability Protocol

Each specified threshold is perturbed individually by -10% , -5% , $+5\%$, and $+10\%$, while the other four remain fixed. This yields 20 perturbation configurations and $20 \times 20 = 400$ instance-level allocation decisions. The perturbed thresholds are reapplied to the recorded descriptor and probe values, and changed allocations are selectively rerun. The protocol measures local decision stability; it is neither a joint parameter search nor a downstream DRC robustness experiment.

The decision trace reports each raw descriptor and a signed normalized distance $(z - \theta)/|\theta|$ to its threshold. For a probed instance, the nearest operator-changing OAT margin is the minimum among an eligible single-gate entry into legacy, entry into XY, and crossing of the probe threshold; analogous active-boundary definitions apply to legacy and XY selections. The signed margin visualizes proximity to a rule boundary.

5. Results

5.1. Comparison with No Inflation and Fixed Legacy

Table 2 reports the main results. As defined in Eq. (25), each Avg. Ratio entry is the arithmetic mean of the experiment/E4 per-design ratios, with E4 normalized to 1.00. This row is descriptive; the paired geometric-mean DRC reductions reported below use Eq. (26).

The primary comparison is against fixed legacy inflation: the complete selector gives a 27.49% paired geometric-mean DRC reduction, with W/T/L = 18/0/2. Relative to the no-inflation baseline, the paired geometric-mean DRC reduction is 73.82%, with W/T/L = 19/0/1.

The DRC improvement is accompanied by higher placement time. The arithmetic means of the matched per-design PT ratios are 0.23 for E0/E4 and 0.49 for E1/E4. The corresponding E0/E4 and E1/E4 RT ratios are 1.38 and 1.15. Thus the baselines use less placement time on average under the experiment/E4 aggregation, whereas their descriptive RT ratios exceed 1.00. DRWL and via counts change only marginally, whereas the dominant change is in DRC.

5.2. Allocation and Branch Ablations

The complete selector allocates one instance to legacy, five to directional XY, six to no-deflation, and eight to scalar. Table 3 reports the fixed-scalar control, static selector, and branch ablations. The DRC Avg. Ratio is 2.62 for E2 and 1.05 for E3, both relative to E4. Relative to the fixed-scalar control, the paired geometric-mean DRC reduction is 21.22%, with W/T/L = 11/8/1. The eight ties are exactly the eight E4 scalar allocations, consistent with the selected path coinciding with the fixed-scalar control on those instances. Adding the probe-conditioned selection path gives a further 4.16% paired geometric-mean DRC reduction relative to E3, with W/T/L = 6/14/0.

Branch knockouts compare RAOP with scalar fallback on the cases assigned to each branch. Removing legacy changes one instance, where retaining the branch reduces DRC by 96.86% relative to E5a. Removing XY changes five instances; the complete selector gives an 8.78% affected-instance geometric-mean DRC reduction, a 2.35% median reduction, and W/T/L = 4/0/1. Removing no-deflation changes six instances; the corresponding geometric-mean DRC reduction is 13.21%, with a 12.53% median reduction and W/T/L = 6/0/0.

5.3. Local Threshold Stability

Table 4 reports threshold decision stability under the OAT perturbations. Overall, 394 of 400 perturbed instance-level allocations remain unchanged (98.50%), and 17 of 20 benchmark instances retain the same allocation under every perturbation.

5.4. Cross-Benchmark Evaluation on ICCAD 2017

Table 5 reports aggregate ratios. Relative to E0, E4 gives a 30.44% paired geometric-mean DRC reduction with W/T/L = 5/0/3. Relative to E1, the paired geometric-mean DRC

reduction is 7.57%, with W/T/L = 3/0/5. Table 6 gives every per-instance DRC count. E4 wins on `edit_dist_1.md1`, `fft_2.md2`, and `fft_a.md3`; E1 wins on the other five instances. The paired geometric mean favors E4 because its three wins are proportionally larger than its five losses; the result is not a majority win.

DRWL and via ratios remain close to one. The E1/E4 PT and RT ratios are 0.442 and 0.895, respectively, indicating additional E4 runtime. E4 selects scalar on six instances and directional XY on two. The cross-benchmark comparison therefore covers two branches under unequal route opportunities and is interpreted descriptively.

5.5. Equal-Route-Opportunity Control

Table 7 reports the equal-route-opportunity control. E4-EQ has a 4.20% lower paired geometric-mean DRC than E1-EQ, but the per-design record is 3/0/5 and the median change is +2.13%. Thus the geometric-mean aggregate favors E4-EQ while the majority and median directions favor E1-EQ. The three E4-EQ wins occur on `edit_dist_1.md1`, `fft_2.md2`, and `fft_a.md2`; Table 8 reports all eight per-design results.

The aggregate direction is sensitive to `edit_dist_1.md1` and `fft_2.md2`; detailed influence analysis is provided in the Supplementary Material.

Secondary outcomes are also mixed. E4-EQ reduces DRWL by 1.76% geometrically, with 3/0/5 and a +0.94% median change. Vias increase by 0.49%, with 1/0/7 and a +1.58% median. Placement time rises by 5.11% geometrically, with 2/0/6 and a +6.79% median change. The geometric aggregate of the per-DEF arithmetic-mean detailed-routing times rises by 29.24%, with 2/0/6 and a +7.58% median change.

6. Discussion

6.1. Mechanistic Interpretation of Operator Selection

The hierarchy assigns specialized responses before using the probe for unresolved states.

The matched comparisons among E2, E3, and E4 suggest that most of the observed improvement over fixed scalar is associated with the static morphology-gated legacy/XY paths, while the probe-conditioned path contributes a smaller additional improvement. In the branch knockouts, every selected non-scalar branch improves the affected-case aggregate relative to scalar fallback. Because these subsets are small and branch-specific, the results do not establish a global operator ranking.

6.2. Practical Trade-offs

The complete selector improves the primary DRC endpoint while consuming more placement time on average than E0 or E1. Fourteen of twenty ISPD 2015 instances undergo the one-step scalar probe. The measured overhead combines this probe with operator-specific placement and global-routing work; component-wise timing was not recorded. The `des_perf_a` row illustrates the trade-off: DRC falls from 50,383 under E1 to 9,843, while RT increases from 393 s to 4,374 s, about 11.1 times higher.

Table 2: Results on the compatibility-processed ISPD 2015 benchmark suite [32, 33]. Instances whose fence-region constraints were removed are marked by †.

Design	E0: No Inflation					E1: Original Legacy					E4: RAOP				
	DRWL/ μm	#DR Vias	#DRCs	PT/s	RT/s	DRWL/ μm	#DR Vias	#DRCs	PT/s	RT/s	DRWL/ μm	#DR Vias	#DRCs	PT/s	RT/s
des.perf.1	1462812	581207	20562	10	1838	1502076	580452	11916	11	2449	1441469	576482	11634	20	1194
des.perf.a†	2322577	561458	29964	7	566	2522313	561220	50383	37	393	2313191	562959	9843	32	4374
des.perf.b†	1779556	549941	15975	7	1406	1839021	514520	2010	9	288	1757112	534220	2026	22	280
edit.dist.a†	5662148	996254	403997	9	2000	5829761	1000486	471427	44	1463	5717317	1009395	373694	45	1716
fft.1	509333	182314	7843	4	997	522235	177012	3683	16	680	509482	181470	3458	22	648
fft.2	592252	184475	8803	4	379	606558	180713	2587	6	900	591873	185075	2398	22	659
fft.a	1092086	189525	5115	4	1056	1129785	180036	1511	5	590	1138284	190119	1348	21	1143
fft.b	1270492	202811	35300	5	710	1322777	205500	16039	13	494	1334636	214909	14132	26	831
matrix_mult.1	2672478	797926	78245	9	1256	2643973	781977	14210	32	2881	2609291	809605	13365	43	3058
matrix_mult.2	2697249	834924	53699	8	1454	2703311	823278	14850	32	4021	2678279	848555	13193	43	5417
matrix_mult.a	3880657	849499	26132	10	4349	4024855	811100	6882	12	4518	3958080	836538	6839	35	5681
matrix_mult.b†	3656645	780504	67258	9	994	3743240	735164	56504	10	839	3783268	786285	44549	27	920
matrix_mult.c†	3711882	802966	26270	8	3288	3761570	749577	7678	10	3355	3571711	771187	6493	25	1094
pci_bridge32.a†	618060	142620	5046	4	1192	640518	134805	2149	4	1204	624196	140333	2001	7	1517
pci_bridge32.b†	968995	143457	2704	4	165	1011454	136659	362	5	105	952674	141787	145	14	89
superblue11.a†	40623289	5614317	1246	42	3978	40621478	5607817	1223	56	3588	41181314	5499516	846	357	4114
superblue12	42776850	10593682	3102568	59	16631	43777837	10318973	14025	368	9325	44462311	10226881	10696	664	10846
superblue14	28001278	4275082	333	30	6510	27999905	4268711	336	46	6536	28057530	4098048	341	325	3497
superblue16.a†	31537174	4579627	4000	31	8964	31535024	4574970	4154	52	11222	31625517	4581793	1478	202	6413
superblue19	20718439	3535785	9411	23	4835	20735257	3509147	5855	56	4115	20739056	3421193	2307	298	2577
Avg. Ratio	1.00	1.01	18.29	0.23	1.38	1.02	0.98	1.56	0.49	1.15	1.00	1.00	1.00	1.00	1.00

† Fence-region constraints were removed during compatibility processing for the nine marked instances.

Notes: E4 is normalized to 1.00; boldface marks the lowest DRC count in each instance row. Detailed-routing time is reported descriptively and is not used as a routability endpoint.

Table 3: Selector Controls and Branch Ablations on the compatibility-processed ISPD 2015 benchmark suite.

Experiment	Method	Avg. Ratio				
		DRWL/ μm	#DR Vias	#DRCs	PT/s	RT/s
E2	Fixed Scalar	0.99	1.01	2.62	1.02	1.00
E3	Static Selector	1.00	1.00	1.05	1.03	0.98
E4	RAOP	1.00	1.00	1.00	1.00	1.00
E5a	w/o Legacy	1.00	1.00	2.54	0.95	1.02
E5b	w/o XY	1.00	1.00	1.03	0.97	1.02
E5c	w/o No-Deflation	1.00	1.00	1.05	1.01	0.99

Avg. Ratio is the arithmetic mean of 20 matched per-instance experiment/E4 ratios. Branch-specific reductions reported in the text use affected instances only.

Table 4: Selector-threshold decision stability of RAOP under one-at-a-time perturbations.

Threshold	Base	-10%	-5%	+5%	+10%	Overall
Mean congestion	0.01	20/20 (100.00%)	20/20 (100.00%)	20/20 (100.00%)	20/20 (100.00%)	80/80 (100.00%)
Top-1% mass	0.2	19/20 (95.00%)	20/20 (100.00%)	20/20 (100.00%)	20/20 (100.00%)	79/80 (98.75%)
Connected hotspot mass	0.1	20/20 (100.00%)	20/20 (100.00%)	20/20 (100.00%)	20/20 (100.00%)	80/80 (100.00%)
H/V imbalance	3	18/20 (90.00%)	19/20 (95.00%)	19/20 (95.00%)	19/20 (95.00%)	75/80 (93.75%)
Probe overflow response	-0.1	20/20 (100.00%)	20/20 (100.00%)	20/20 (100.00%)	20/20 (100.00%)	80/80 (100.00%)

Table 5: Cross-benchmark evaluation on eight compatibility-processed ICCAD 2017 instances. Metric columns report the arithmetic mean of matched per-instance E/E4 ratios; larger values indicate a larger lower-is-better metric than E4. DRC reduction is the paired geometric-mean reduction of E4 relative to the named reference.

Reference	DRWL	DR Vias	DRCs	PT	RT	E4 DRC reduction	W/T/L
E0 No inflation	1.020	1.005	1.824	0.203	1.040	30.44%	5/0/3
E1 Fixed legacy	1.031	0.998	1.144	0.442	0.895	7.57%	3/0/5
E4 RAOP	1.000	1.000	1.000	1.000	1.000	—	—

The equal-route-opportunity control separates routing opportunity from runtime. Matching route opportunity does not equal matching runtime; route count alone is therefore insufficient for

a cost comparison.

RAOP requires a first-route congestion field, directional demand/capacity, a complete placement-state snapshot for roll-

Table 6: Per-instance DRC counts on compatibility-processed ICCAD 2017. E4/E1 below one denotes an E4 win; all eight instances are shown.

Design	E0	E1	E4	E4/E1	E4 vs. E1
des_perf_b.md1	22,883	17,273	22,908	1.326	Loss
des_perf_b.md2	21,221	16,357	21,482	1.313	Loss
edit_dist_1.md1	189,343	43,198	31,577	0.731	Win
edit_dist_a.md2	570,249	357,514	428,571	1.199	Loss
fft_2.md2	16,958	11,331	7,554	0.667	Win
fft_a.md2	10,056	9,723	9,846	1.013	Loss
fft_a.md3	9,867	19,392	9,917	0.511	Win
pci_bridge32_a.md1	5,916	5,761	5,826	1.011	Loss

Table 7: Equal-route-opportunity control on eight compatibility-processed ICCAD 2017 instances. Ratios are paired geometric E4-EQ/E1-EQ aggregates; lower is better. Median is computed over the eight per-design percentage changes.

Metric	Geometric ratio	Change	Median change	W/T/L
DRC	0.958013	-4.198711%	+2.128495%	3/0/5
DRWL	0.982446	-1.755392%	+0.938068%	3/0/5
DR vias	1.004875	+0.487536%	+1.580226%	1/0/7
Placement time	1.051135	+5.113501%	+6.786848%	2/0/6
Per-DEF mean DR time	1.292371	+29.237078%	+7.575607%	2/0/6

Table 8: All per-design DRC outcomes in the equal-route-opportunity control. A ratio below one denotes an E4-EQ win.

Design	E1-EQ DRC	E4-EQ DRC	E4-EQ/E1-EQ	Change	E4-EQ result
des_perf_b.md1	17,785	23,215	1.305313	+30.5313%	Loss
des_perf_b.md2	16,357	21,349	1.305190	+30.5190%	Loss
edit_dist_1.md1	48,787	34,527	0.707709	-29.2291%	Win
edit_dist_a.md2	393,168	433,578	1.102780	+10.2780%	Loss
fft_2.md2	11,499	7,711	0.670580	-32.9420%	Win
fft_a.md2	12,803	9,768	0.762946	-23.7054%	Win
fft_a.md3	9,891	10,097	1.020827	+2.0827%	Loss
pci_bridge32_a.md1	5,795	5,921	1.021743	+2.1743%	Loss

back, and repeated global routing. A fixed-policy bypass remains useful when turnaround is constrained. Porting the selector to another placer, congestion estimator, or detailed router requires revalidating the descriptor distributions, rollback semantics, and quality-runtime trade-off.

6.3. Validity and Generalization Boundary

The ISPD 2015 study uses the same processed design family that informed the threshold scales and is therefore a development-family evaluation. It exercises all four response policies and supports the complete-system and ablation analyses. The two ICCAD 2017 campaigns use eight processed designs and activate scalar and XY only, providing a narrower test of the frozen selector on a second design family.

The ICCAD results apply to compatibility-processed inputs and to the scalar and XY branches only. Repeat-level time dispersion cannot be assessed. Evaluation on constraint-preserving inputs, other routers and technologies, and external cases that exercise every response policy remains future work.

7. Conclusion

RAOP selects among routing-response policies with different actuation and admissibility semantics using first-route mor-

phology and, for unresolved states, one rollback-capable probe. On 20 compatibility-processed ISPD 2015 instances, it reduces paired geometric-mean DRC by 27.49% relative to fixed legacy inflation. The matched controls and branch ablations show contributions from both the morphology-gated and probe-conditioned paths within this development family.

On compatibility-processed ICCAD 2017, the frozen selector is evaluated under matched route opportunity. The DRC aggregate favors RAOP, while per-design outcomes and runtime costs remain design dependent. Broader evaluation is needed to establish generality.

Data and code availability

Releasable supporting artifacts are publicly available at <https://github.com/beimeibaiyang/raop>. The repository includes machine-readable result tables for all reported comparisons, the selector decision trace and threshold-stability records, public input-transformation audits, the equal-route protocol and opportunity-count audit, a code-only implementation diff, manuscript snapshots, checksums, and a script that recomputes the reported aggregates. The complete source code is not publicly released. Additional implementation materials

and non-public experimental records can be provided to the editors and reviewers upon request during peer review, subject to applicable benchmark, software, and institutional restrictions. Detailed-routing reproduction requires licensed Cadence Innovus access. Benchmark inputs, generated and routed DEFs, per-repeat reports, and raw Innovus logs are not publicly available.

Ethics declaration

This software and benchmark study involves no human participants, personal data, animals, or clinical intervention; institutional ethics approval was not applicable.

CRedit authorship contribution statement

jjh: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **rainy**: Validation, Investigation, Writing – review & editing. **fj**: Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

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